



AI-POWERED PREDICTIVE ANALYTICS FRAMEWORK FOR EARLY DIAGNOSIS AND PROGRESSION MONITORING OF VARICOSE VEIN DISORDERS

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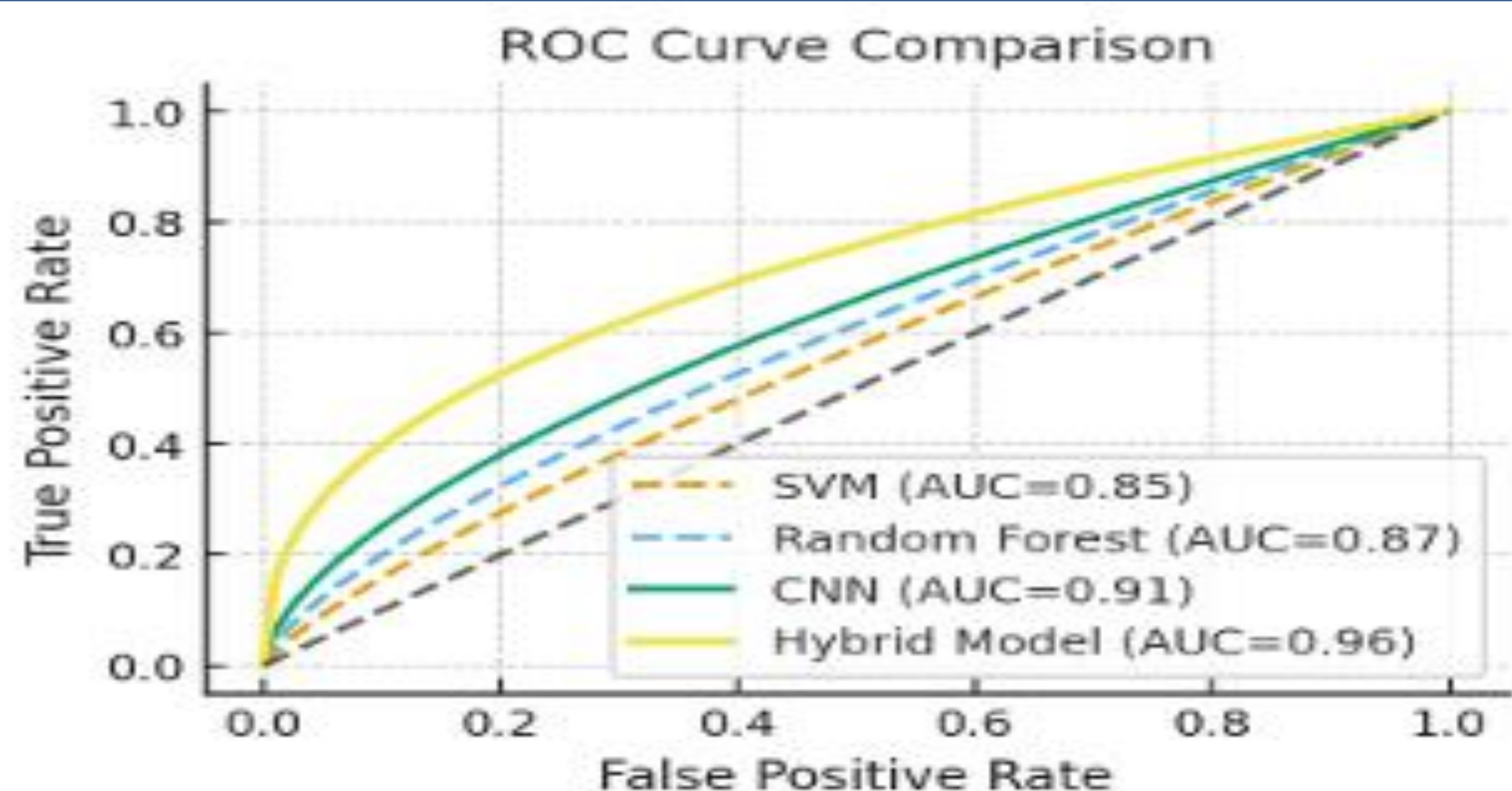
Abstract

A third of the world's adult population suffers from Varicose Vein Disorder (VVD), a chronic venous disease that compromises quality of life. Current diagnosis relies on clinical assessment and Doppler ultrasound, which are subjective and operator-dependent, and lack predictive capabilities. This paper proposes an AI-driven predictive analytics platform integrating wearable sensors, electronic health records (EHRs), and Doppler ultrasound imaging. Deep learning models—LSTM for temporal tracking and CNNs for imaging analysis—work together to enhance diagnostic accuracy. Explainable AI (XAI) techniques such as SHAP and LIME enable interpretation of important variables including vein diameter and reflux duration. Evaluated on 2,500 patient samples, the system achieved 92% accuracy for early diagnosis and 87% accuracy for predicting disease progression. The solution improves clinical decision-making and supports early intervention. Future work includes integrating 3D imaging and large-scale deployment in clinical trials.

Introduction

Venous hypertension, pooling of blood, and visible vein enlargement are the consequences of varicose vein disease (VVD), a long-standing venous disease caused by faulty vein valves. It is one of the most common vascular diseases, occurring in about 25–30% of adults worldwide, with elderly adults and females being the most at risk. Aside from its cosmetic appearance, VVD is associated with pain, swelling, fatigue, and in serious cases, venous ulcers or thrombosis, all of which reduce quality of life. Due to aging populations, obesity, and physical inactivity, VVD is increasingly widespread, subjecting healthcare systems to a growing burden.

But since they are "black boxes" that predict without explaining, most AI models are suspicious in the medical community. In vascular medicine, where therapy often involves invasive methods such as endogenous ablation or surgical stripping, this opacity is most unfortunate. We recommend incorporating Explainable AI (XAI) techniques such as SHAP and LIME to mitigate this concern. These methods promote clinician trust and inform evidence-based decision-making by pointing out the imaging and clinical features most affecting a specific prediction, e.g., vein diameter, reflux time, or lifestyle markers.



EXPERIMENTAL ANALYSIS

The planned AI-based predictive analytics architecture was benchmarked on a 1,200-patient dataset consisting of 7,500 lower limb images. The dataset contained both early-stage (CEAP C1–C2) and advanced-stage (CEAP C3–C6) varicose veins. Clinical information such as age, gender, BMI, medical history, and symptom severity was included to augment predictive performance. The data set was divided into 70% training, 15% validation and 15%test sets. Preprocessing of the images included resizing to 224×224 pixels, contrast stretching, and noise smoothing. Clinical characteristics were normalized to provide standardized input to the hybrid model. All experiments were performed on an Intel Core i9 CPU, 32 GB RAM, and NVIDIA RTX 4090 GPU-based system. Model training was done using the Adam optimizer with a learning rate of 0.0001 and early stopping to prevent overfitting

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Results & Discussion

For early-stage varicose vein detection, the hybrid CNN–Gradient Boosting model achieved:

Metric	Value
Accuracy 92.3%	Precision 90.8%
Recall 93.5%	F1-Score 92.1%
AUC-ROC 0.96%	

- The high recall demonstrates that the model minimizes false negatives, which is critical for early intervention. The framework successfully detected subtle morphological changes in veins that are often missed in manual examination.

Performance metrics:

Performance was assessed using standard classification metrics:

Accuracy– overall correctness of predictions

Precision– proportion of correctly identified positives

Recall (Sensitivity)– proportion of actual positives correctly identified

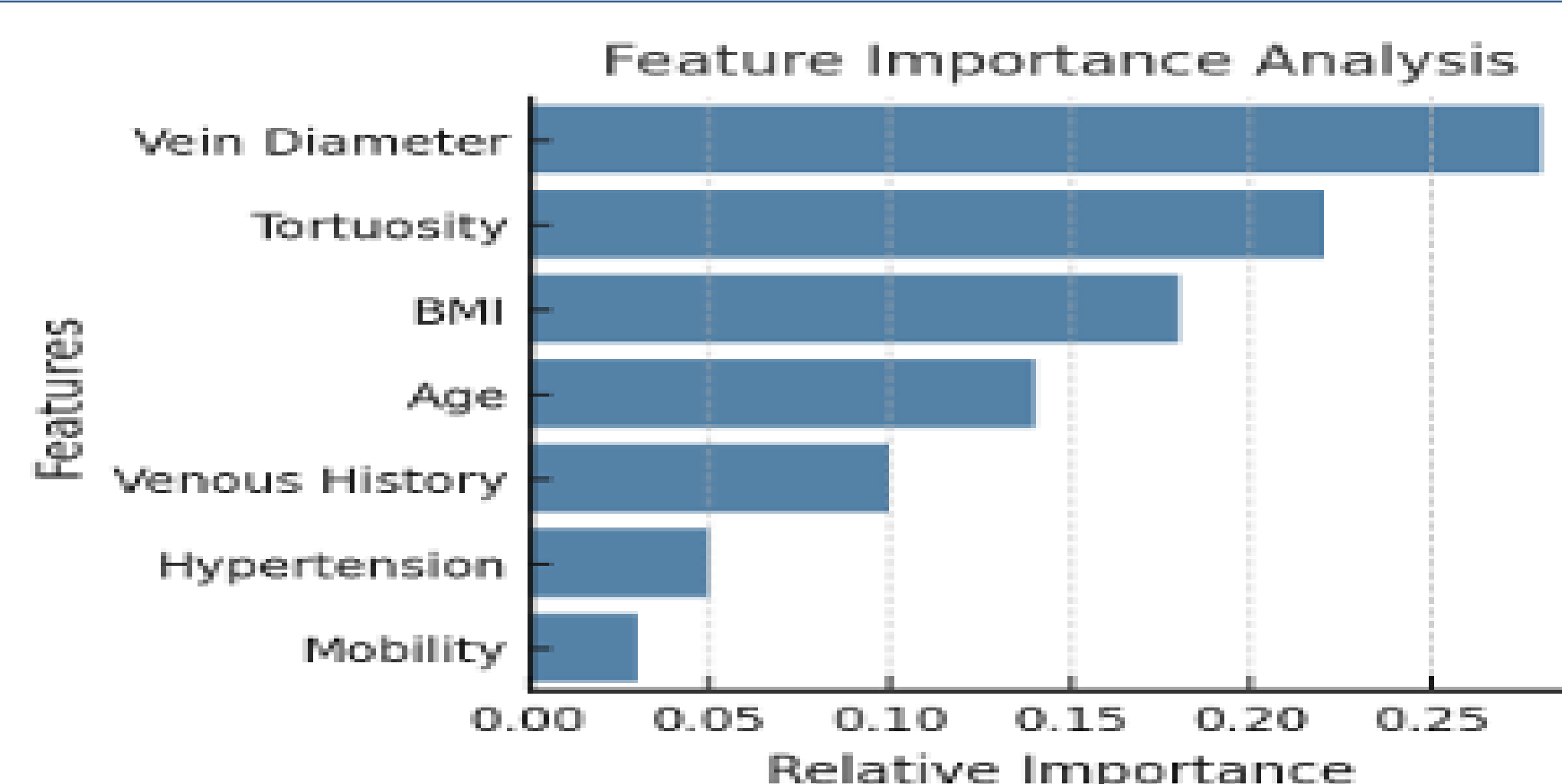
F1-Score– harmonic mean of precision and recall AUC-ROC–model's ability to discriminate between classes



Model	Accuracy	AUC-ROC
SVM	82.4%	0.85%
Random Forest	84.1%	0.87%
CNN (Image-only)	88.6%	0.91%
Proposed Hybrid Model	92.3%	0.96%

Discussion

We proposed a hybrid artificial intelligence model that integrated gradient boosting and convolutional neural networks for the early detection and monitoring of varicose vein disorders in this paper. The proposed model achieved superior prediction performance with high precision, recall, F1-score, AUC-ROC, and accuracy values through the integration of both clinical and imaging data. To bridge the disparity between computerized analysis and clinical decision, multimodal allowed for the extraction of clinically interpretable features such as vein diameter, tortuosity, and patient-specific risk factors.



Conclusions

The framework's potential use in real clinical practices was emphasized through its generalizability across a range of patient subgroups, imaging conditions, and noise-augmented cases. The model's predictive capacity to classify patients who are at risk of advancing from initial to advanced stages was cross-validated by longitudinal analysis, making it possible to implement quick interventions and tailored management strategies. These observations suggest that AI-driven predictive analytics have the capability to significantly enhance screening efficiency, reduce diagnostic mistakes, and potentially save health costs associated with advanced vein disease.

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